

Complex Analysis solution

5.2 #6

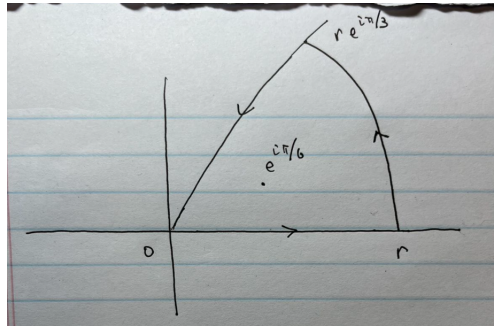
Proof. Let $\gamma_r = [0, x] + re^{it} - se^{i\pi/3}$ for $x \in [0, r]$, $t \in [0, \pi/3]$ and $s \in [0, r]$ as drawn in the picture. The line integral along γ_r is

$$\begin{aligned} \int_{\gamma_r} \frac{1}{1+z^6} dz &= \int_0^r \frac{1}{1+x^6} dx + \int_0^{\pi/3} \frac{1}{1+(re^{it})^6} rie^{it} dt - \int_0^r \frac{1}{1+(se^{i\pi/3})^6} e^{i\pi/3} ds \\ &= (1 - e^{i\pi/3}) \int_0^r \frac{1}{1+x^6} dx + \int_0^{\pi/3} \frac{1}{1+r^6 e^{i6t}} rie^{it} dt. \end{aligned}$$

By Residue theorem $\int_{\gamma_r} \frac{1}{1+z^6} dz = 2\pi i \text{Res}(f, e^{i\pi/6}) = -\frac{\pi i}{3} e^{i\pi/6}$. When r large enough, it is clear $|1 + r^6 e^{i6t}| \geq r^6 - 1 \geq r^6/2$. Then by ML-estimate, we have

$$\left| \int_0^{\pi/3} \frac{1}{1+r^6 e^{i6t}} rie^{it} dt \right| \leq \int_0^{\pi/3} \frac{1}{|1+r^6 e^{i6t}|} r dt \leq \frac{2}{r^5} \frac{\pi r}{3} \rightarrow 0 \text{ as } r \rightarrow \infty.$$

Taking $r \rightarrow \infty$, we have $-\frac{\pi i}{3} e^{i\pi/6} = \lim_{r \rightarrow \infty} \int_{\gamma_r} f(z) dz = (1 - e^{i\pi/3}) \int_0^\infty \frac{1}{1+x^6} dx$ □

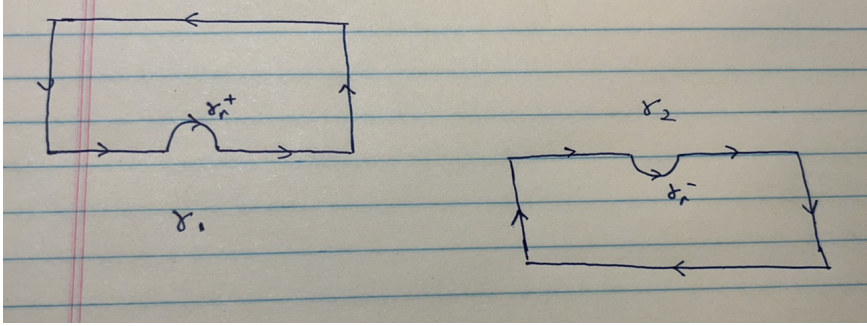


5.3 #6

Proof. We use Theorem 5.3.8 and argument in Theorem 5.3.2. Note

$$\int_{-R}^{-r} \frac{\sin z}{z} dz + \int_r^R \frac{\sin z}{z} dz = \left(\int_{-R}^{-r} \frac{1}{2iz} e^{iz(1-t)} dz + \int_r^R \frac{1}{2iz} e^{iz(1-t)} dz \right) - \left(\int_{-R}^{-r} \frac{1}{2iz} e^{iz(-1-t)} dz + \int_r^R \frac{1}{2iz} e^{iz(-1-t)} dz \right)$$

Denote $I = \int_{-R}^{-r} \frac{1}{2iz} e^{iz(1-t)} dz + \int_r^R \frac{1}{2iz} e^{iz(1-t)} dz$ and $II = \int_{-R}^{-r} \frac{1}{2iz} e^{iz(-1-t)} dz + \int_r^R \frac{1}{2iz} e^{iz(-1-t)} dz$. Let γ_1 the indented path in Figure 5.3.3 and γ_2 be the reflection of γ_1 with respect to x -axis. We draw the paths γ_i 's below with orientations labelled.



We use different paths to evaluate different integrals. There are 5 cases.

1. When $1 - t < 0$, then $-1 - t < 0$. As $R \rightarrow \infty, r \rightarrow 0$, we claim

$$I = \int_{\gamma_2} \frac{1}{2iz} e^{iz(1-t)} dz \quad II = \int_{\gamma_2} \frac{1}{2iz} e^{iz(-1-t)} dz.$$

Indeed the integral along the vertical and horizontal lines are zero since

$$\lim_{\substack{z \rightarrow \infty \\ y = \text{Im}(z) < 0}} \left| \frac{1}{2iz} e^{iz(1-t)} \right| \leq \frac{1}{2|z|} e^{-y(1-t)} \rightarrow 0$$

as $\text{Im}(z) = y \rightarrow -\infty$. The same argument shows $\lim_{\substack{z \rightarrow \infty \\ y = \text{Im}(z) < 0}} \left| \frac{1}{2iz} e^{iz(-1-t)} \right| = 0$ since $-1 - t < 0$.

Then when $R \rightarrow \infty, r \rightarrow 0$ by Theorem 5.3.8,

$$I = -\gamma_r^- = -i(\pi - 2\pi) \text{Res}\left(\frac{1}{2iz} e^{iz(1-t)}, 0\right) = \frac{\pi}{2}$$

$$II = -\gamma_r^- = -i(\pi - 2\pi) \text{Res}\left(\frac{1}{2iz} e^{iz(-1-t)}, 0\right) = \frac{\pi}{2}.$$

So $I - II = 0$ which implies $\hat{f}(t) = 0$.

2. When $1 - t > 0$ and $-1 - t > 0$. This amounts to $t < -1$. Then

$$I = \int_{\gamma_1} \frac{1}{2iz} e^{iz(1-t)} dz \quad II = \int_{\gamma_1} \frac{1}{2iz} e^{iz(-1-t)} dz.$$

Here we note $\lim_{\substack{z \rightarrow \infty \\ y = \text{Im}(z) < 0}} \left| \frac{1}{2iz} e^{iz(-1-t)} \right| \leq \frac{1}{2|z|} e^{-y(-1-t)} \rightarrow 0$ when $\text{Im}(z) = y \rightarrow \infty$. The same

argument shows $\lim_{\substack{z \rightarrow \infty \\ y = \text{Im}(z) < 0}} \left| \frac{1}{2iz} e^{iz(1-t)} \right| = 0$. Again when $R \rightarrow \infty, r \rightarrow 0$, by Theorem 5.3.8,

$$\begin{aligned} I &= -\gamma_r^+ = -i(\pi - 0) \text{Res}\left(\frac{1}{2iz} e^{iz(1-t)}, 0\right) = -\frac{\pi}{2} \\ II &= -\gamma_r^+ = -i(\pi - 0) \text{Res}\left(\frac{1}{2iz} e^{iz(-1-t)}, 0\right) = -\frac{\pi}{2} \end{aligned}$$

So $I - II = 0$ which implies $\hat{f}(t) = 0$.

3. When $1 - t > 0$ and $-1 - t < 0$, then $-1 < t < 1$. By argument above

$$\begin{aligned} I &= -\gamma_r^+ = -i(\pi - 0) \text{Res}\left(\frac{1}{2iz} e^{iz(1-t)}, 0\right) = -\frac{\pi}{2} \\ II &= -\gamma_r^- = -i(\pi - 2\pi) \text{Res}\left(\frac{1}{2iz} e^{iz(-1-t)}, 0\right) = \frac{\pi}{2} \end{aligned}$$

So $I - II = -\pi$ which implies $\hat{f}(t) = -\pi$.

4. Suppose $t = 1$. Then $I = 0$ and $II = \int_{\gamma_2} \frac{1}{2iz} e^{iz(-1-t)} dz = \frac{\pi}{2}$. It follows $\hat{f}(1) = -\pi/2$.

5. Suppose $t = -1$. Then $I = -\frac{\pi}{2}$ and $II = 0$. It follows $\hat{f}(1) = -\pi/2$.

□

5.3 #7

Proof. By definition $\hat{f}(t) = \int_{-\infty}^{\infty} f(x)e^{-itx}dx$. By integration by parts, we have

$$\hat{f}(t) = -\frac{1}{it}e^{-itx}f(x)\Bigg|_{-\infty}^{\infty} - \int_{-\infty}^{\infty} -\frac{1}{it}e^{-itx}f'(x)dx = -\frac{1}{it}e^{-itx}f(x)\Bigg|_{-\infty}^{\infty} + \frac{1}{it}\hat{f}'(t).$$

Since $x, t \in \mathbb{R}$, by assumption $\lim_{x \rightarrow \infty} |e^{-itx}f(x)| = \lim_{x \rightarrow \infty} |e^{-itx}||f(x)| = \lim_{x \rightarrow \infty} |f(x)| = 0$, which implies

$\lim_{x \rightarrow \infty} e^{-itx}f(x) = 0$. Similarly $\lim_{x \rightarrow -\infty} e^{-itx}f(x) = 0$. Thus $-\frac{1}{it}e^{-itx}f(x)\Bigg|_{-\infty}^{\infty} = 0$ and $\hat{f}(t) = \frac{1}{it}\hat{f}'(t)$. $\square$